Scanning electron microscopy & energy dispersive X-ray spectroscopy of multi-layered, mixed-composition glass nanoparticles

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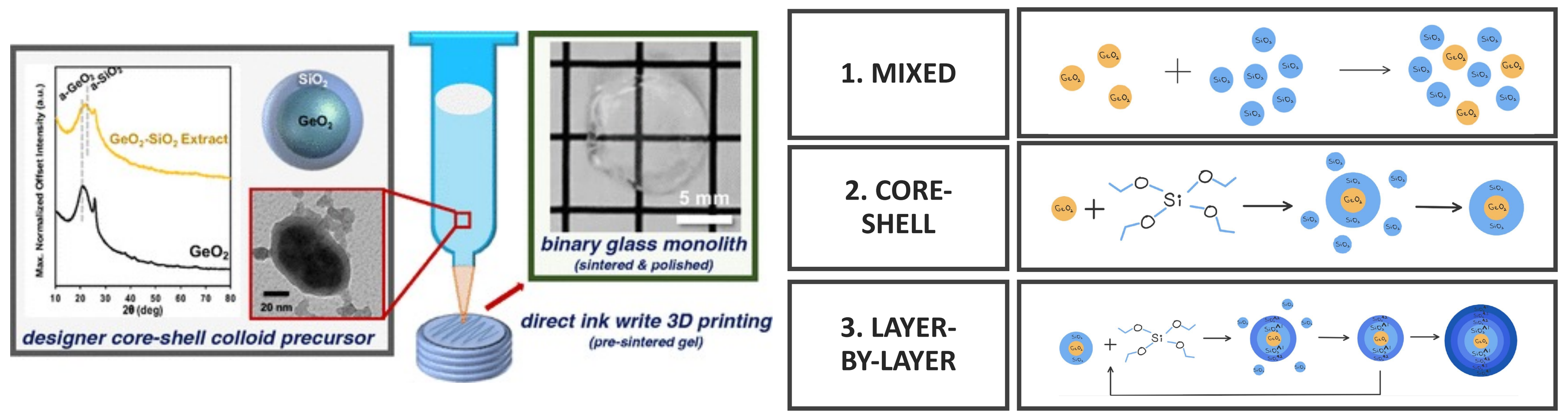
Background & Motivation

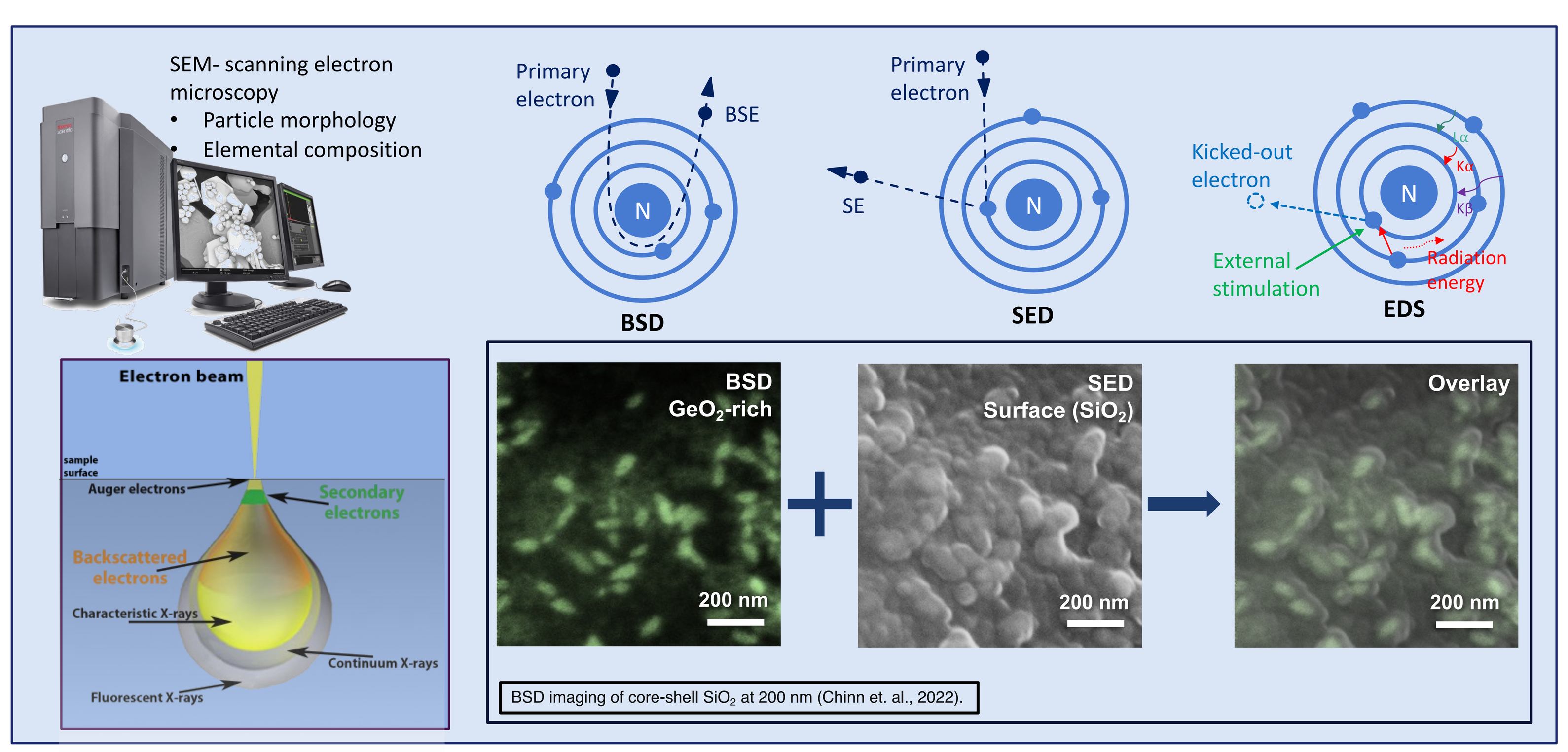
Germania-silica nanoparticles (a.k.a., colloids) have been proven to be useful for additive manufacturing optical glass materials. Theoretically, these nanoparticles will enable control of the glass composition at the nanometer scale. Previous research has focused on (1) silica and germania colloids prepared individually and then mixed[1] and (2) silica-encapsulated germania colloids[2]; however, both exhibit limitations related to germania domain control. Research on newly engineered nanoparticles suggest that the layer-by-layer method provides increased control over composition and domain size, and improved glass network homogeneity.



Objectives

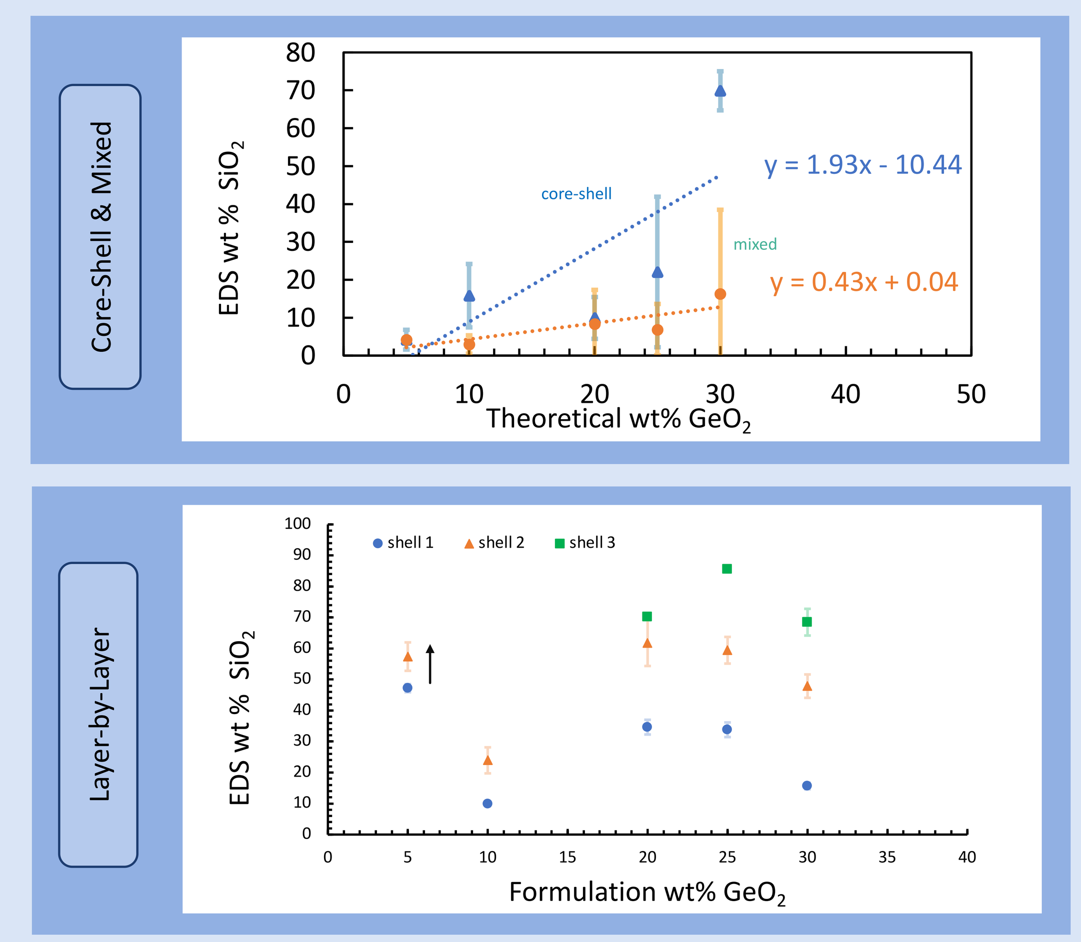
1. Fabricate and compare layer-by-layer GeO_2-SiO_2 nanoparticles (NPs) at small range of GeO_2 weight percent formulations
2. Analyze and compare the microstructure and elemental composition of germania-silica, layer-by-layer nanoparticles in various stages of processing using scanning electron microscopy (SEM) with energy-dispersive X-ray spectroscopy (EDS)

Methodology



Results

Previous EDS Study

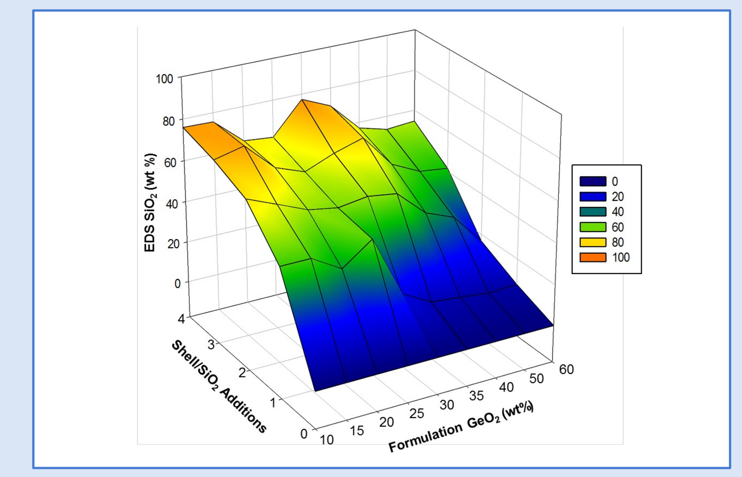


Layer-by-Layer EDS Study

Actual Weight Percent SiO_2

	Core	Shell 1	Shell 2	Shell 3	Shell 4
10%	0.0	45.3 ± 6.2	64.6 ± 4.2	71.5 ± 6.2	76.0 ± 4.2
15%	0.0	45.4 ± 12.2	58.7 ± 5.7	74.9 ± 5.2	75.6 ± 7.1
20%	0.0	36.9 ± 11.3	52.7 ± 4.1	61.4 ± 9.7	63.3 ± 17.5
25%	0.0	47.4 ± 25.7	50.3 ± 13.1	56.1 ± 6.4	61.9 ± 8.4
30%	0.0	17.6 ± 7.5	52.4 ± 3.8	61.9 ± 8.4	77.4 ± 4
35%	0.0	9.9 ± 3.5	50.4 ± 16.1	66.2 ± 9.7	71.2 ± 3.9
40%	0.0	8.4 ± 2	37.6 ± 1	50.1 ± 2.7	57.2 ± 13.5
50%	0.0	6.8 ± 2.8	32.0 ± 17	43.3 ± 3.2	53.4 ± 1.8
60%	0.0	7.4 ± 5	16.7 ± 6.5	41.6 ± 5.8	54.5 ± 7.5

Figure 1. Heat map of actual weight percent SiO_2 at various weight percent formulations and SiO_2 shells



Sintering Study

Figure 5. Average Diameter of particles after each addition of SiO_2 at all weight percent GeO_2 formulation ranges

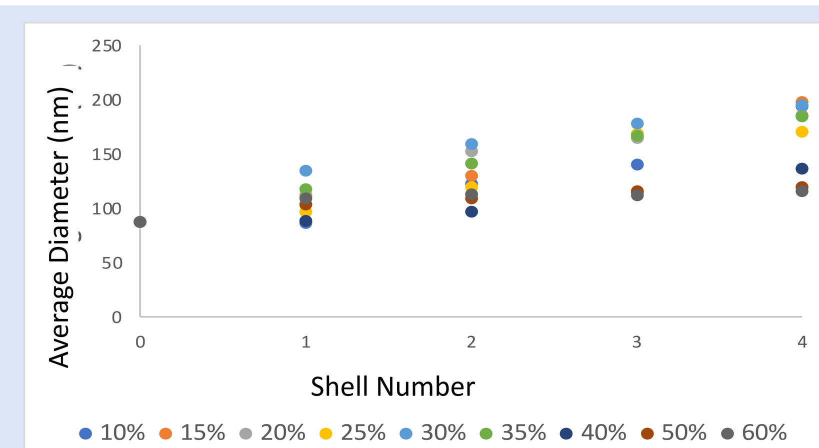
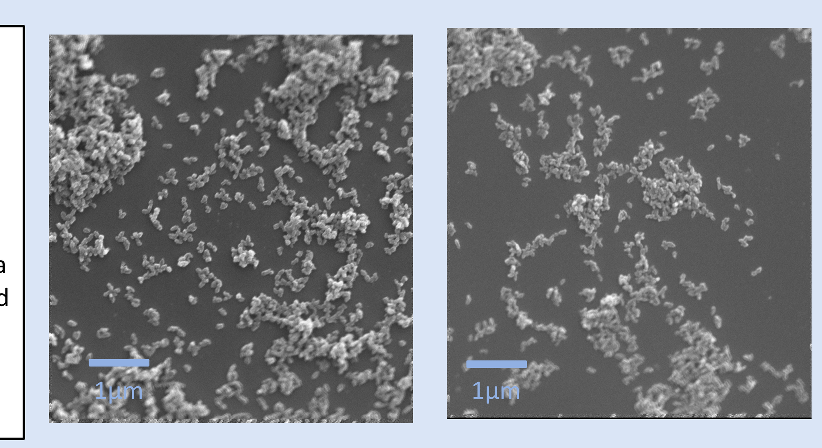


Figure 6. SED images taken at 50,000 times magnification, used for FIJI Image analysis



Sintering Study

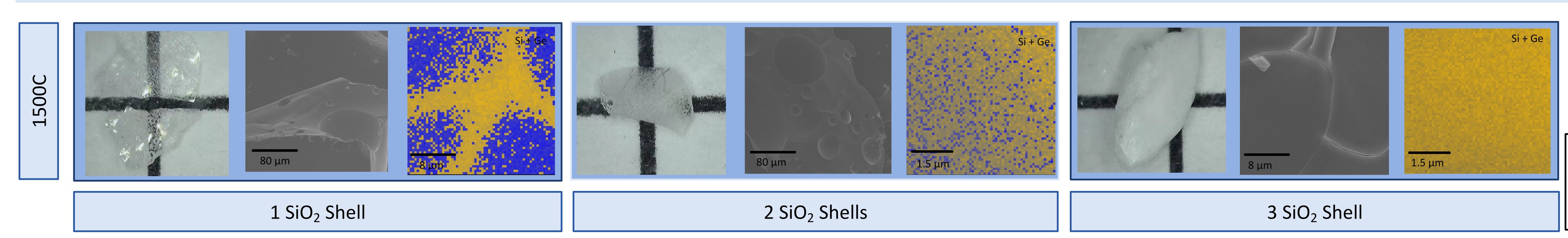


Figure 7.

Figure 9. EDS on 30 wt% GeO_2 formulations taken when particles were suspended, compressed into a pellet, and heated at 1300, 1400, and 1500 degrees Celsius.

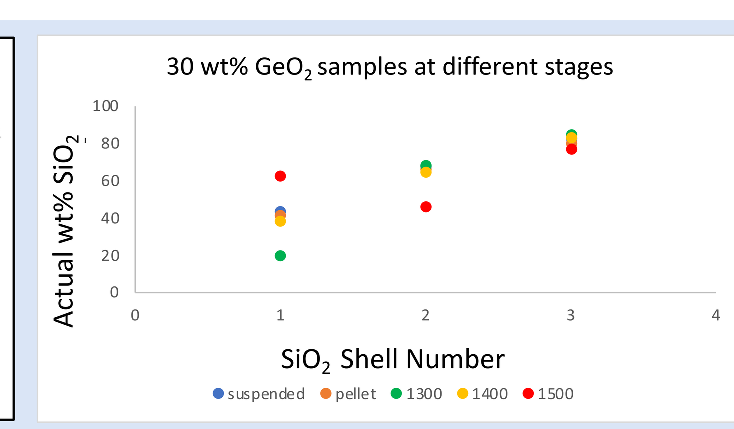
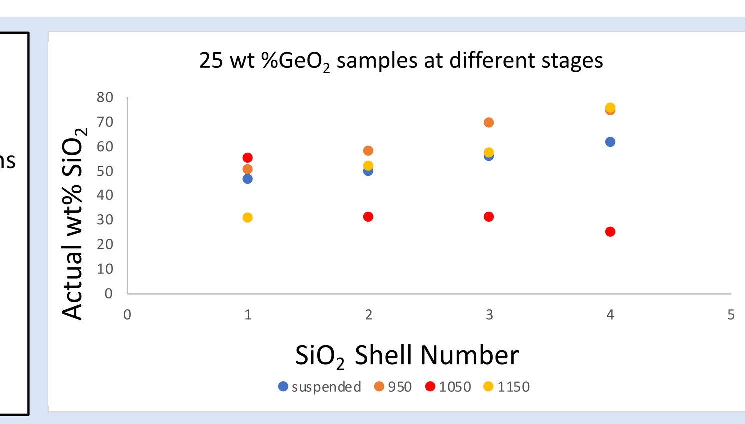


Figure 10. EDS on 25 wt% GeO_2 formulations taken when particles were suspended,, and heated at 950, 1050, and 1150 degrees Celsius.



Lower Temperature Sintering Study

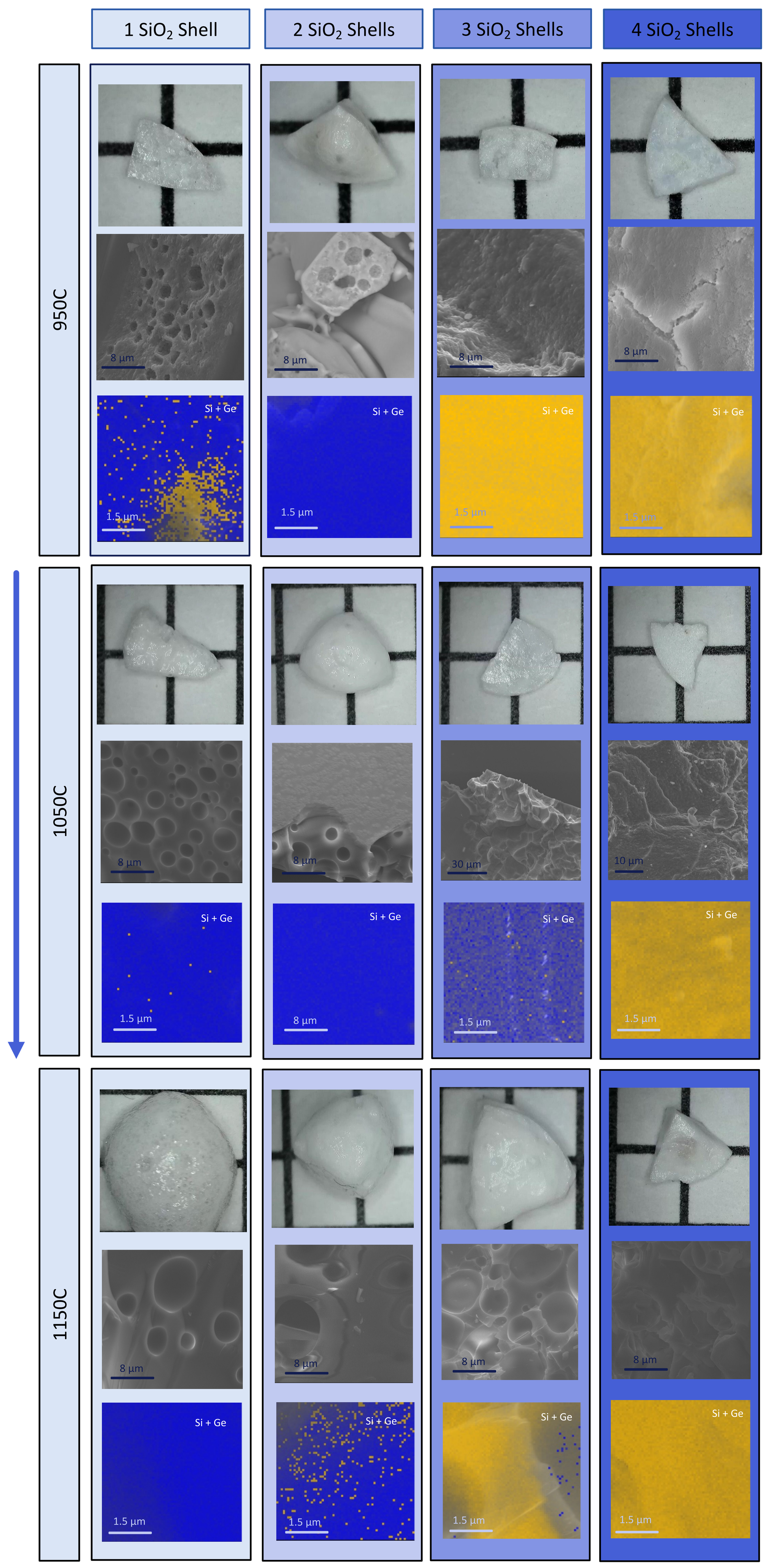


Figure 7(left) & 8(above). For each sample from left to right (top to bottom for figure 8): image of heated sample, SED image of heated sample surface, overlay of EDS SiO_2 and GeO_2 mapping. For figure 8 Samples are shown increasing in the amount of SiO_2 shells from left to right and increasing in temperature going down. Figure 7 shows 30 wt% GeO_2 formulations and figure 8 show 25 wt% GeO_2 formulations

Conclusion

Layer-by-layer NPs–

- Adding SiO_2 to GeO_2 core particles was consistent from layer to layer, diameters steadily increased, as shown by SEM for all weight percent formulations tested; however, growth was less drastic for 50 and 60 weight percent formulations.
- EDS confirms an increase in weight percent SiO_2 with the addition of each layer for all weight percent formulations tested.
- Overall, results show that layer-by-layer particles provide increased control of composition and homogeneity at a wide range of formulations and prove the utility of the layer-by-layer method for obtaining specific SiO_2 and GeO_2 weight percent compositions.

Layer-by-layer sintering-

- Samples with a lower amounts of shells melted to transparency more readily, as expected due to the lower melting point of Germania.
- Less homogeneity was observed in samples at 1500 degrees. This could be due to diffusion during melting. Lower temperatures maintained homogeneity and prevented possible diffusion.
- Overall, we have made significant progress in producing and processing materials, and they have behaved predictably under viscous sintering.

Future Objectives

1. Replicate studies to ensure results are consistent
2. Replicate sintering studies on more weight percent GeO_2 formulations, temperatures, and times

References

1. Sasan et. al. *ACS App. Mater. Interfaces.* **2020**, 12 (5), 6736-6741. doi: 10.1021/acsami.9b21136
2. Chinn, A., et al. *ACS Omega* **2022** 7 (20), 17492-17500 DOI: 10.1021/acsomega.2c02292

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